

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter-2023

Course: B. Tech

Semester :IV

Branch : Electronics and Tele-communication Engineering

Subject Code & Name: BTBSC406_Y18 Numerical Methods & Computer Programming

Max Marks: 60

Date:29/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Find the absolute error & relative error in $\sqrt{6} + \sqrt{7} + \sqrt{8}$ correct to 4 significant digits.	CO 02	6
B)	Define errors. How many different types of errors occur while doing numerical computations? How they can be avoided?	CO 01	6
C)	If $X^2 - e^{-X} = 0$, find real root by Newton Raphson method	CO 02	6
Q.2	Solve Any Two of the following.		12
A)	Solve the following equations $27x+6y-z = 85$ $X+y+54z = 110$ $6x+15y+2z = 72$ By using Gauss Seidal method	CO 03	6
B)	Solve the following by Gauss elimination method $2x+y+z = 10$ $3x+2y+3z = 18$ $X+4y+9z = 16$	CO 04	6
C)	Solve the following equations by LU decomposition (Factorization method) $2x+3y+z = 9$ $X+2y+3z = 6$ $3x+y+2z = 8$	CO 03	6
Q. 3	Solve Any Two of the following.		12
A)	Apply Lagrange's formula inversely to obtain a root of the equation $f(x)=0$, given that $f(30)=-30, f(34)=-13, f(38)=3, f(42)=18$	CO 05	6

B)	The following values of $y = f(x)$ are given X : 10 15 20 F(x) : 1754 2648 3564 Find the value of x for $y = 3000$ by iterative method	CO 03	6
C)	Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using i) Trapezoidal rule ii) Simpson's 1/3rd rule iii) Simpson's 3/8th rule	CO 04	6
Q.4	Solve Any Two of the following.		12
A)	Apply Runge kutta 4 th order method to find approximate value of y for x = 0.2 in steps of 0.1, if $dy/dx = x+y^2$ given that $y=1$ where $x=0$.		6
B)	Given $dy/dx = y-x / y+x$ with initial condition $y=1$ at $x=0$, find y for $x=0.1$ by using Euler's method		6
C)	Given the values X : 5 7 11 13 17 F(X) : 150 392 1452 2366 5202 Evaluate $f(9)$ using Newton's divided difference formula.	CO 05	6
Q. 5	Solve Any Two of the following.		12
A)	Explain following terms : i) Data hiding ii) Encapsulation iii) Polymorphism	CO 06, CO 07	6
B)	Explain the advantages of OOP? List features of OOP.	CO 06	6
C)	Differentiate between procedural and object oriented approach.	CO 07	6

*** End ***